

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Bachelor of Computer Applications (B.C.A.)

Semester - I University Examination March - 2018

CA 111 Fundamental Concepts of Programming Language

Date: 27.03.2018, Tuesday

Time: 10.00 a.m. to 01.00 p.m.

Maximum Marks: 70

Instructions:

1. Attempt both sections in separate answer books.
2. The figure to the right indicates marks.
3. Make a suitable assumption when necessary.

SECTION - I

Q-1 Do as directed.

[A] State whether the following statements are [True/False]. [06]

1. Object files are generated by the linker.
2. In Repeat Until, the sequence will be performed at least once.
3. There must be only one start and one stop symbol in a flowchart.
4. All the identifiers must be written in lowercase (small) letters.
5. A keyword can be written in uppercase, lowercase or combination of both.
6. The input of a linker is an executable or runnable file.

[B] Fill in the blanks. [05]

1. _____ provides a blueprint for writing a program to solve a particular problem.
a) Algorithm b) Program c) Software d) None of these
2. A _____ is any statement that may evaluate either to a true value or a false value.
a) Loop b) Condition c) Variable d) None of these
3. _____ are identifiers whose values does not change.
a) Variables b) Arrays c) Constants d) Functions
4. Which file contains the program statements?
a) Header File b) Object File c) Source code File d) Executable File
5. Execution of the program begins with _____.
a) Comment b) Header files c) main() d) Function

Q-2 Answer the following questions.

[A] Explain *structure of C* program. [06]

[B] Explain all the *methods of constant variable declaration* with example. [06]

OR

Q-2 Answer the following questions.

[A] Explain with diagram *compilation and execution* of C program. [06]

[B] Write a note on *symbols used in flowchart*. [06]

Q-3 Answer the following questions. [Any Three]

[A] Discuss *advantages and disadvantages* of a flowchart.

[B] Draw a *flowchart* to add two numbers.

[C] Explain various *keywords* used to write a *pseudocode*.

[D] Explain all the *characteristics* of C programming language.

[E] Write an *algorithm* to print first 10 natural numbers.

SECTION - II

Q-4 Do as directed.

[A] State whether the following statements are [True/False]. [06]

1. A function must send back any values to the calling function.
2. In a ternary operator, true part is considered after : symbol.
3. We may omit the size at the time of array initialization.
4. The break statement is an optional.
5. switch cannot be nested.
6. Use of the parameter names in function declaration is optional.

[B] Fill in the blanks. [05]

1. C has built-in multi way decision statement known as _____.
a) switch b) for c) if d) while
2. The _____ statement is used to branch unconditionally from one point to another in a program.
a) break b) goto c) transfer d) label
3. The return type must be _____ if no values are returned.
a) int b) void c) float d) char
4. Array is a _____ data type.
a) basic b) fundamental c) derived d) core
5. In _____ the data type is promoted from a lower level to a higher level.
a) Type Casting b) Explicit Type Conversion c) Loose Typing d) Implicit Type Conversion

Q-5 Answer the following questions.

[A] Which are the elements of *user defined function*? Explain in detail. [06]

[B] Explain *switch* statement with proper example. [06]

OR

Q-5 Answer the following questions.

[A] Write a note on *array* and explain it with proper example. [06]

[B] Write a program to find *summation* of n numbers using an *array*. [06]

Q-6 Answer the following questions. [Any Three]

[12]

[A] What is called *loop*? Explain *types of loop* with example.

[B] Write a C program to *swap two numbers* using *function*.

[C] Write down rules for operator *precedence* and *associativity*.

[D] Write a note on *operators* used in C with proper example.

[E] Write short note: *Implicit Type Conversion*.